# LOAD BALANCING

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## PRACTICAL ON ALB

**1>create a server-1**

1. create instance
2. name-------server1
3. machine---amazon linux
4. AZ------1a
5. sg------ssh,http,https
6. user data

   #!/bin/bash
   yum update -y
   yum install httpd -y
   echo "<h1> server1 "$HOSTNAME" </h1>" > /var/www/html/index.html
   systemctl start httpd.service
   systemctl enable httpd.service
7. launch a instance

**2>create a server-2**

1. create instance
2. name---server2
3. machine---amazon linux
4. AZ------1a
5. sg------ssh,http,https
6. user data

   #!/bin/bash
   yum update -y
   yum install httpd -y
   echo "<h1> server2 "$HOSTNAME" </h1>" > /var/www/html/index.html
   systemctl start httpd.service
   systemctl enable httpd.service
7. launch a instance

**3>create a target group**

1. create a target group
2. target type------instance
3. target name----tgserver

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4. protocol----------http
5. health check path-----/
6. register targets-------- server1 & server2
7. include as a pending below
8. create a target group

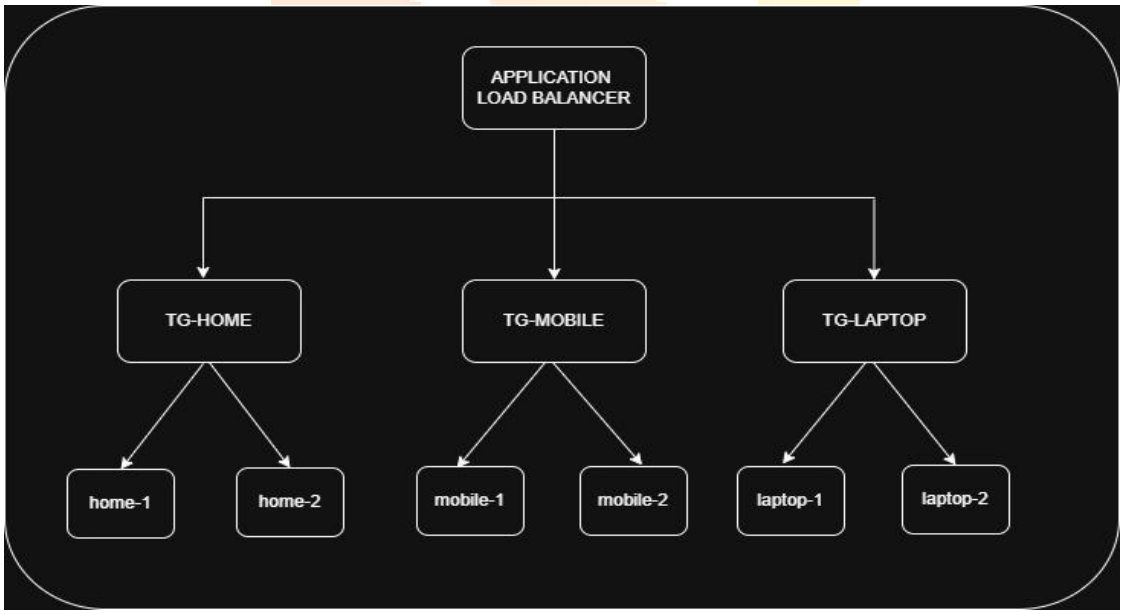
**4> create a application load balancer**

1. application load balancer
2. name------ALB
3. scheme----internet facing (publically)
4. mapping---1a,1b
5. forward to ----> tgserver
6. create a load balancer

5> (dns of ALB) -------> hit on web

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## PRACTICAL ON ALB (PATH BASED ROUTING)




**1>create a home instance**

1. name-------home
2. count ------ 2
3. machine---amazon linux
4. sg------ssh,http,https
5. user data
   #!/bin/bash

yum update -y
yum install httpd -y
echo "<h1> these is my home instance "$HOSTNAME" </h1>" >
/var/www/html/index.html
systemctl start httpd.service
systemctl enable httpd.service

6. launch a instance

**2>create a mobile instance**

1. name-------mobile
2. count ------2
3. machine---amazon linux
4. sg------ssh,http,https
5. user data
   #!/bin/bash
   Sudo yum update -y
   Sudo yum install httpd -y
   Sudo mkdir -p /var/www/html/mobile
   echo "<h1> These is my mobile instance of "$HOSTNAME" </h1>" >
   /var/www/html/mobile/index.html
   systemctl start httpd.service
   systemctl enable httpd.service
6. launch a instance

**3>create a laptop instance**

1. name-------laptop
2. count ------2
3. machine---amazon linux
4. sg------ssh,http,https
5. user data
   #!/bin/bash
   Sudo yum update -y
   Sudo yum install httpd -y
   Sudo mkdir -p /var/www/html/laptop
   echo "<h1> These is my laptop instance of "$HOSTNAME" </h1>" >
   /var/www/html/laptop/index.html
   systemctl start httpd.service
   systemctl enable httpd.service
6. launch a instance

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**3>Create A Target Group (Home)**

1. create a target group
2. target type------instance
3. target name----tg-home
4. protocol---------http
5. health check path-----/
6. register targets-------- home1 & home2
7. include as a pending below
8. create a target group

**4>Create A Target Group (Mobile)**

1. create a target group
2. target type------instance
3. target name----tg-mobile
4. protocol----------http
5. health check path-----/mobile/
6. register targets-------- mobile1 & mobile2
7. include as a pending below
8. create a target group

**5>Create A Target Group (Laptop)**

1. create a target group
2. target type------instance
3. target name----tg-laptop
4. protocol----------http
5. health check path-----/laptop/
6. register targets-------- laptop1 & laptop2
7. include as a pending below
8. create a target group

**6> Create A Application Load Balancer**

1. application load balancer
2. name-------ALB
3. scheme----internet facing
4. mapping---1a,1b,1c
5. forward to ----> tg-home
6. create a load balancer

**7>Add rules in the Load Balancer**

1. click on load balancer
2. click on http 80

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3. manage rules ------> add rules
4. condition -----> path -----> /mobile/
5. forward to tg ----> tg-mobile

**8>Add rules in the Load Balancer**

1. click on load balancer
2. click on http 80
3. manage rules ------> add rules
4. condition -----> path -----> /laptop/
5. forward to tg ----> tg-laptop

**5> Paste the domain name**

1. (dns of ALB) -------> get the home page
2. dns of ALB/mobile/ ---------> get the mobile page
3. dns of ALB/laptop/ ----------> get the laptop page

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## CONFIGURE NGINX AS A LOAD BALANCER

**1>create a server-1**

1. name------- server1
2. machine---amazon linux
3. sg------ssh,http,https
4. user data

```
#!/bin/bash
yum update -y
yum install httpd -y
echo "<h1> server1 "$HOSTNAME" </h1>" > /var/www/html/index.html
systemctl start httpd.service
systemctl enable httpd.service
```

5. launch a instance

**2>create a server-2**

1. name------- server2
2. machine---amazon linux
3. sg------ssh,http,https
4. user data

```
#!/bin/bash
Sudo yum update -y
Sudo yum install httpd -y
echo "<h1> server2 "$HOSTNAME" </h1>" > /var/www/html/index.html
```

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systemctl start httpd.service
systemctl enable httpd.service

5. launch a instance

**3>create a server-3**

1. name------- load balancer
2. machine---amazon linux
3. sg------ssh,http,https
4. launch a instance

**4>Access the load-balancer server**

1. yum install nginx -y
2. systemctl start nginx
3. systemctl enable nginx
4. cp /etc/nginx/nginx.conf /etc/nginx/nginx.conf.backup
5. vim /etc/nginx/nginx.conf

Replace the http section & add the following inside the existing http {} block

```
http {
   upstream backend_servers {
      server 172.31.11.101;          #private_ip of server-1
      server 172.31.12.102;          #private_ip of server-2
   }
   server {
      listen 80;
      location / {
         proxy_pass http://backend_servers;
         proxy_set_header Host $host;
         proxy_set_header X-Real-IP $remote_addr;
         proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
      }
   }
}
```

6. nginx -t
7. systemctl restart nginx

**5>Access the Load Balancer server**

1. (public_ip_of_LB_server)
2. Refresh the page several times
3. You get
   1. server-1
   2. server-2

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## LOAD BALANCER WITH THE RULE 40-40-20 WEIGHTED POLICY

**1.create a instance**

1. sudo apt update -y
2. sudo apt install nginx -y
3. sudo systemctl start nginx

**1.create a instance**

1. sudo apt update -y
2. sudo apt install apache2 -y
3. sudo systemctl start apache2

**3.create a instance**

1. sudo apt update -y
2. apt install openjdk-21-jdk -y
3. wget https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.23/bin/apache-tomcat-11.0.23.tar.gz
4. tar -xvf apache-tomcat-11.0.23.tar.gz
5. cd apache-tomcat-11.0.23
6. cd bin/
7. ./catalina.sh start

**4.create a target group (nginx)**

1. name------tg-nginx
2. type-------instance
3. path---------/
4. register-------nginx
5. create a target group

**5.create a target group (apache2)**

1. name------tg-apache2
2. type-------instance
3. path---------/
4. register-------apache2
5. create a target group

**6. create a target group (tomcat)**

1. name------tg-tomcat
2. type-------instance
3. path---------/
4. register-------tomcat-----8080----include as pending below
5. create a target group

**7.create a load balancer**

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1. name----------ALB
2. default action-------tg-nginx
3. create a load balancer

**8.click on created ALB**

1. select listener------click on listener
2. action----------edit listener
3. add target group-----tg-apache2 and tg-tomcat
4. add the weighted policy
    1. tg-nginx----------4----------40%
    2. tg-apache2------4 ----------40%
    3. tg-tomcat--------2----------20%

**9.click on application load balancer that you created**

1. copy the dns and paste

**10.The traffic is on nginx 40%, apache2 40% and in tomcat about 20%**

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**TASK:**

1. Path-Based Routing using Application Load Balancer (ALB) (home/mobile/laptop/watch)
2. Path-Based Routing (Home/Mobile/Laptop) with Custom Ports (8001/8002/8003)
3. Configure LAMP server
4. ALB Health Check Failure Handling
5. ALB Weighted Routing with (40%-40%-20%) Rule
6. Nginx as a load balancer

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